Ch 7.4: Cubic splines

Lecture 23 - CMSE 381

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Fri, Nov 1, 2024

Announcements

Last time:

- 7.2 Step functions
- 7.3 Basis functions

This lecture:

- 7.4 Cubic splines

Announcements:

- Homework # 6 is due Sunday
- Projects

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
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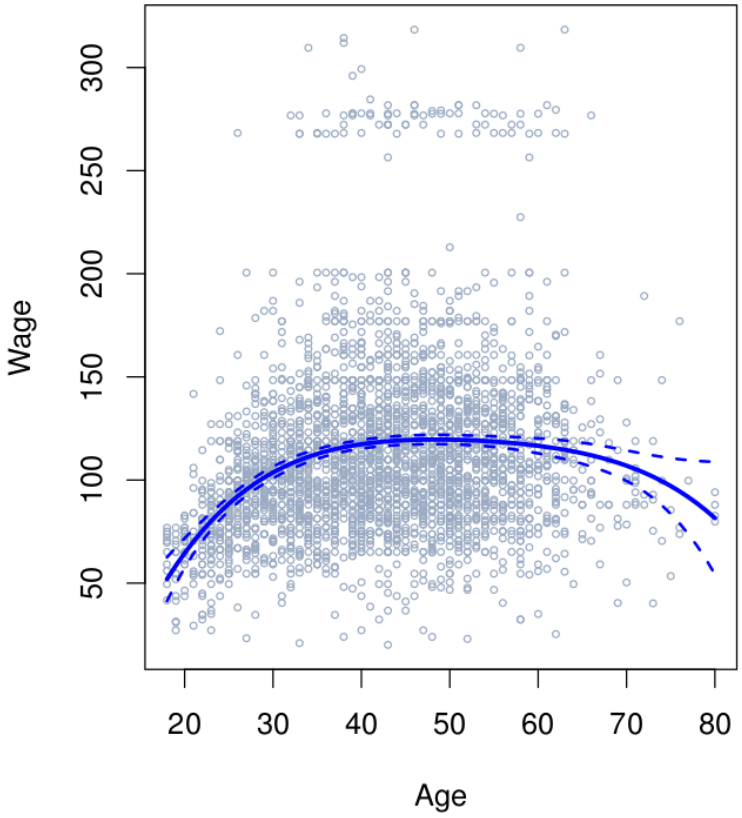
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		

Section 1

Last time

Polynomial regression

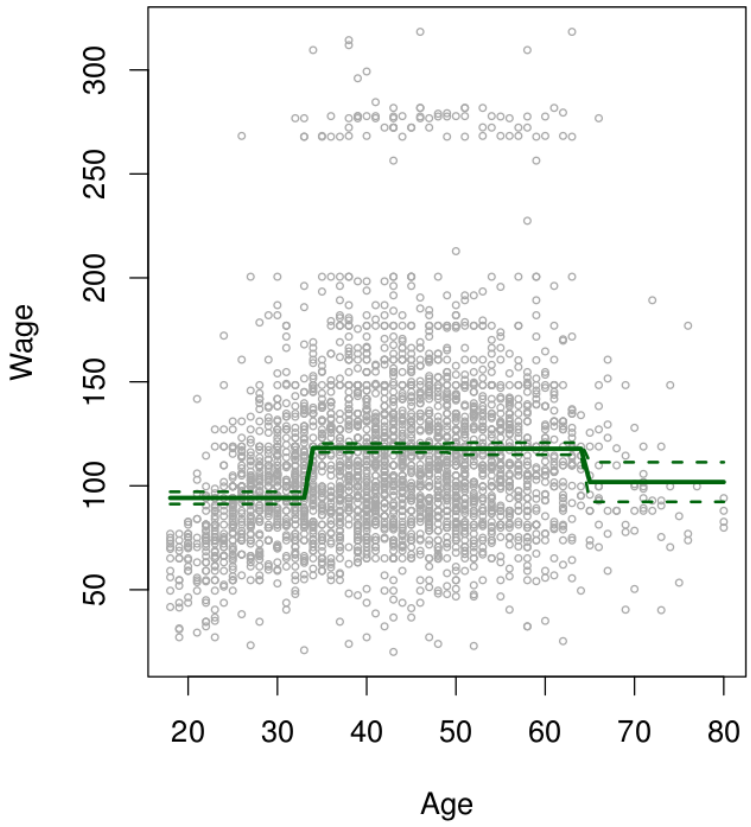
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$



Step function regression

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \leq X < c_2), \\
C_2(X) &= I(c_2 \leq X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\
C_K(X) &= I(c_K \leq X),
\end{aligned}
$$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

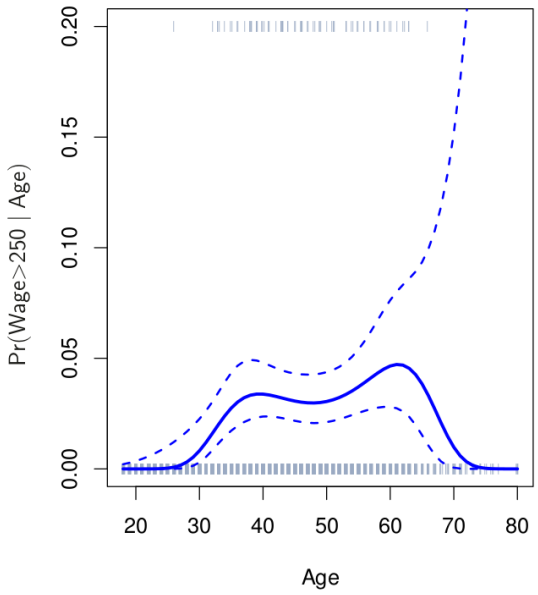


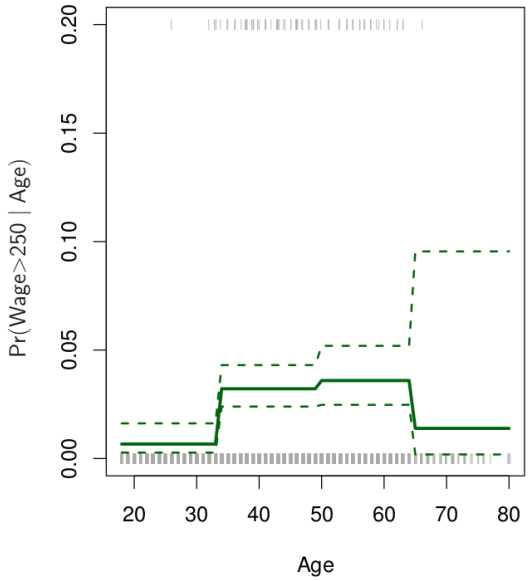
Classification version

$$\Pr(y_i > 250 \mid x_i) =$$

$$\frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$

$$\frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$





Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 \underline{b_1(x_i)} + \beta_2 \underline{b_2(x_i)} + \cdots + \beta_K \underline{b_K(x_i)} + \varepsilon_i$$

$b_1, \ldots, b_K$ are basis functions

Section 2

Regression Splines

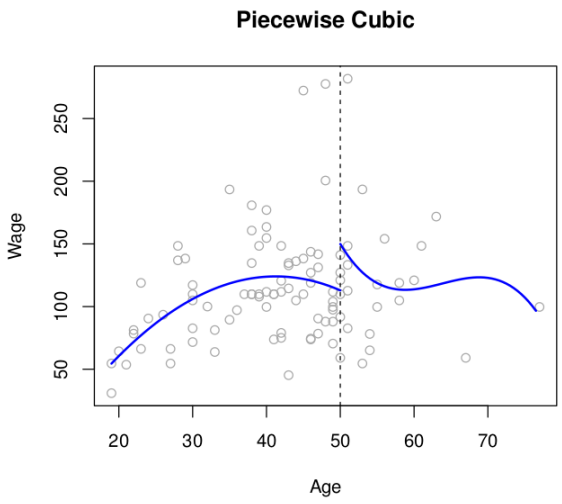
Piecewise polynomials

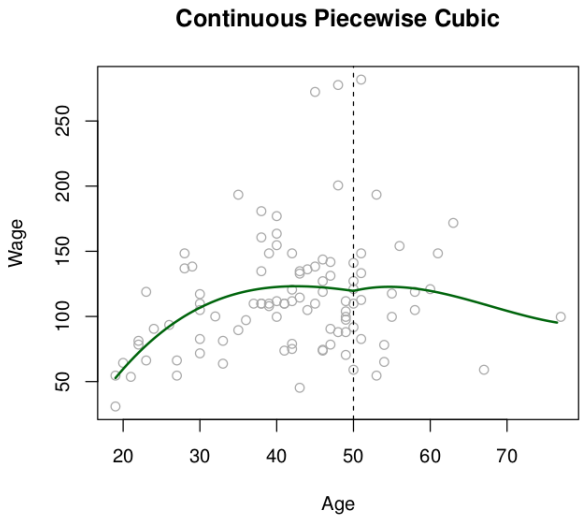
- Fit a polynomial regression

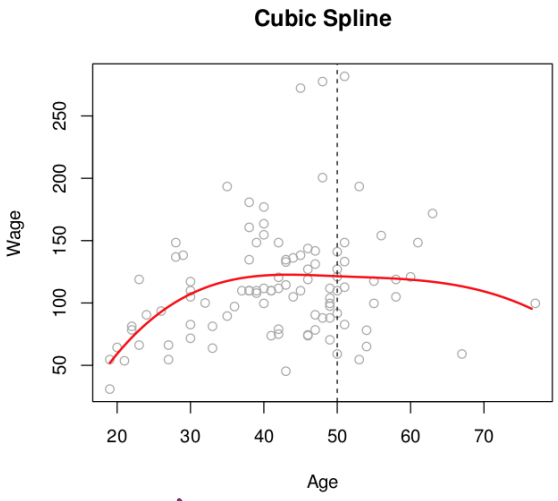
  $$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.

Building up to cubic splines







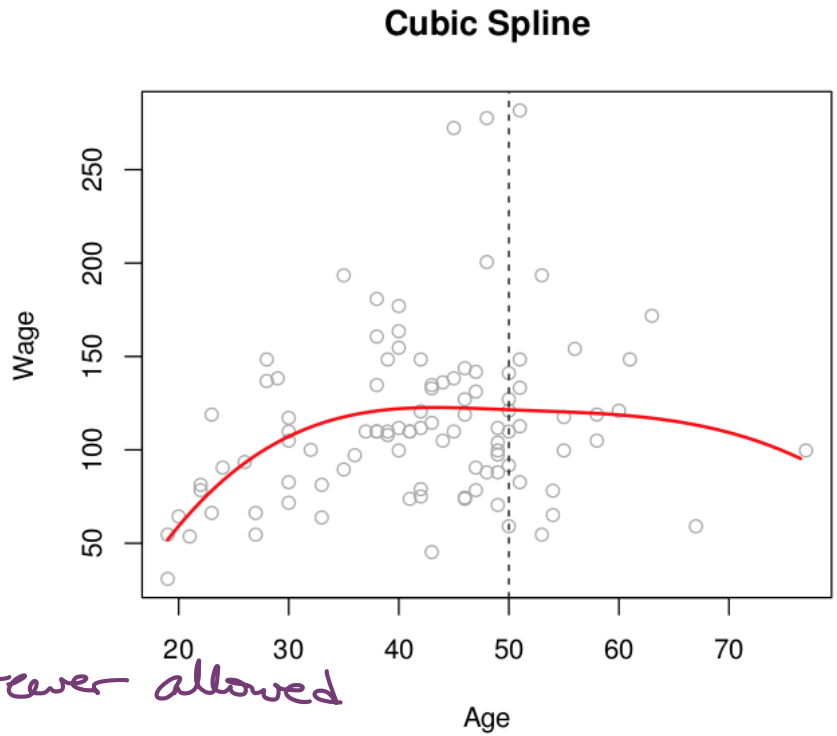
deg 3 piecewise polynomial

continuous 1st, 2nd deriv are cont.

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

Cubic splines: degrees of freedom

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 + \beta_3^1 x^3 & x < c \\ \beta_0^2 + \beta_1^2 x + \beta_2^2 x^2 + \beta_3^2 x^3 & x > c \end{cases}$$



8 degrees of freedom ⇒ piecewise cubic

Requiring cont, 1st, 2nd cont ⇒ 3 fewer allowed

one knot ⇒ 8 − 3 = 5 deg of freedom

Generally K knots ⇒ ~~4+K~~ 3+K degrees of freedom

Spline basis representation

$\leftarrow b_1, \ldots, b_{K+3}$

Want to pick b_i so that we represent a cubic spline with K knots as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

Warning: Many ways to do this!

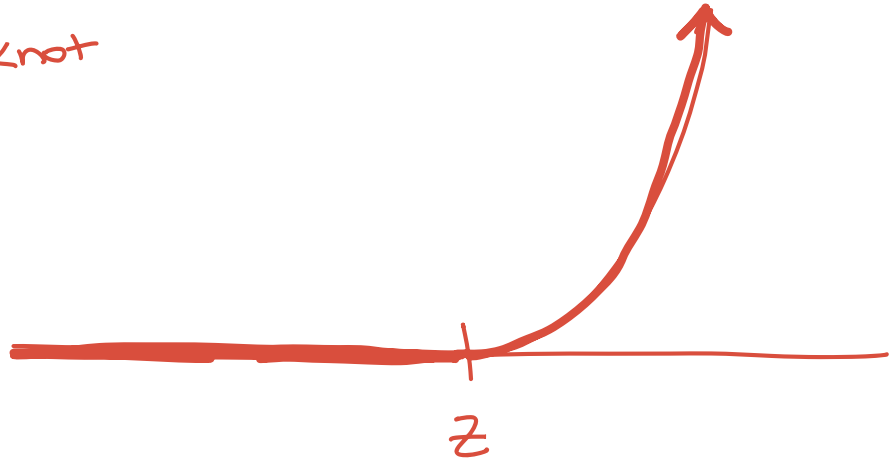
Version 1: Truncated power basis function

$$h(x,z) = (x-z)_+^3 = \begin{cases} (x-z)^3 & \text{if } x > z \\ 0 & else \end{cases}$$

knot

$0 + 0x + 0x^2 + 0x^3$

Desmos link: https://www.desmos.com/calculator/esucuulbgj



This is a cubic spline!

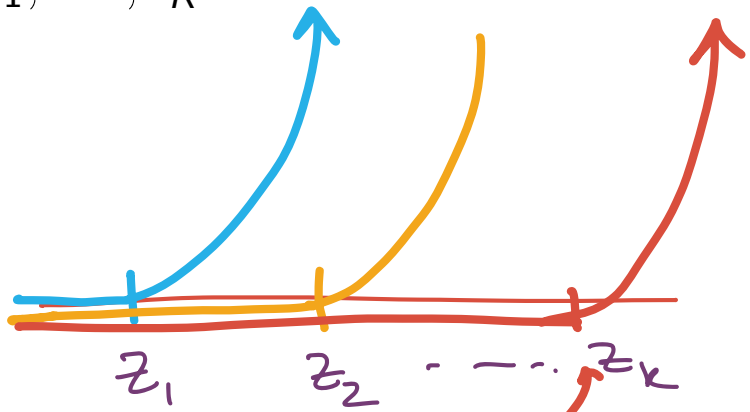
piecewise cubic

cont, 1st, 2nd deriv cont

The (book) basis for cubic splines

Given knots at $z_1, \cdots, z_K$

- b_1 X
- b_2 X^2
- b_3 X^3
- $h(X, z_1)$
- $h(X, z_2)$
- $\vdots$
- $h(X, z_K)$

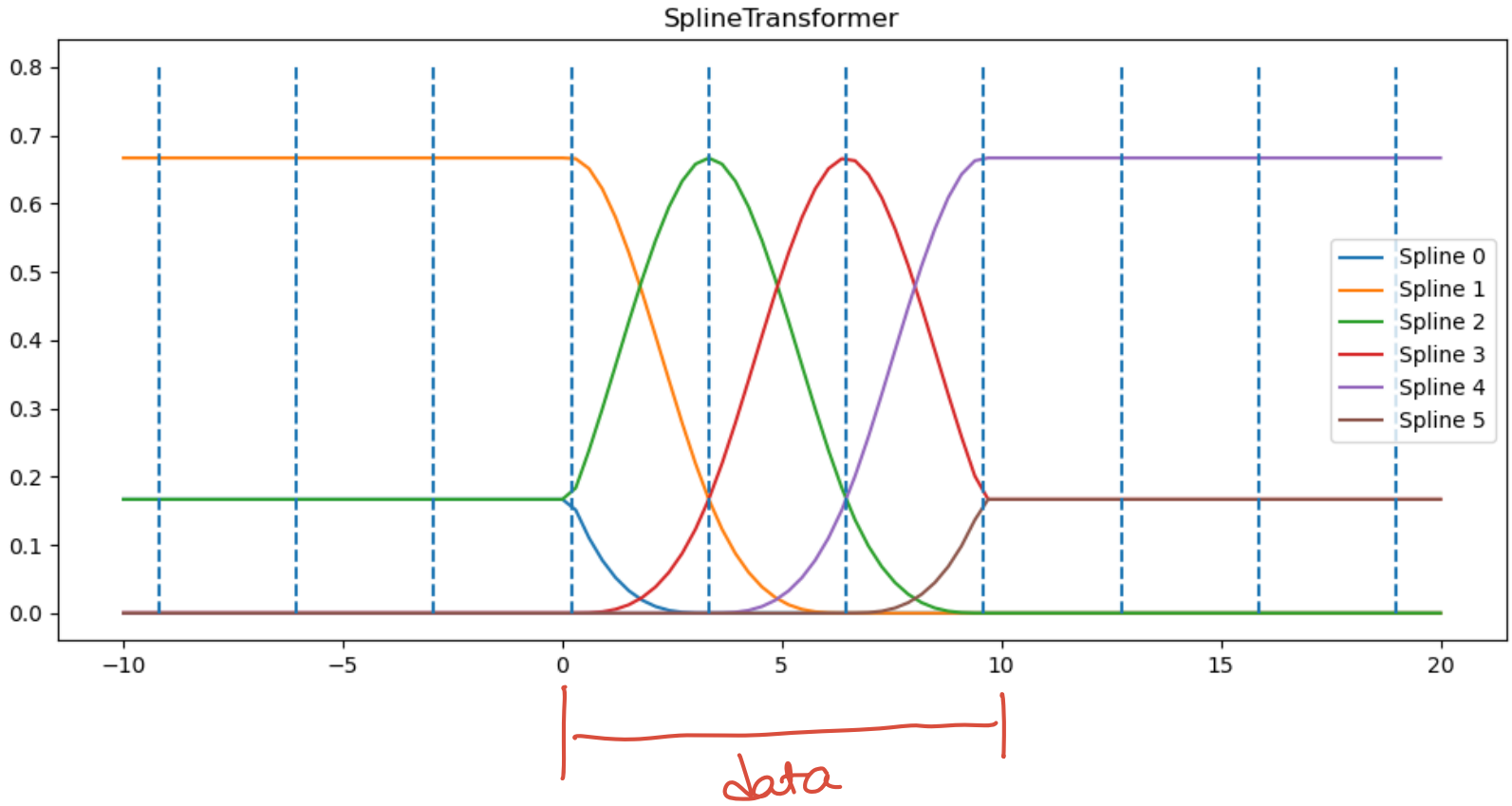


Promise:

Result is a cubic spline

$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 h(X, z_1) + \beta_5 h(X, z_2) + \cdots + \beta_{k+3} h(X, z_K)$$

Version 2: B-spline basis function



Coding example

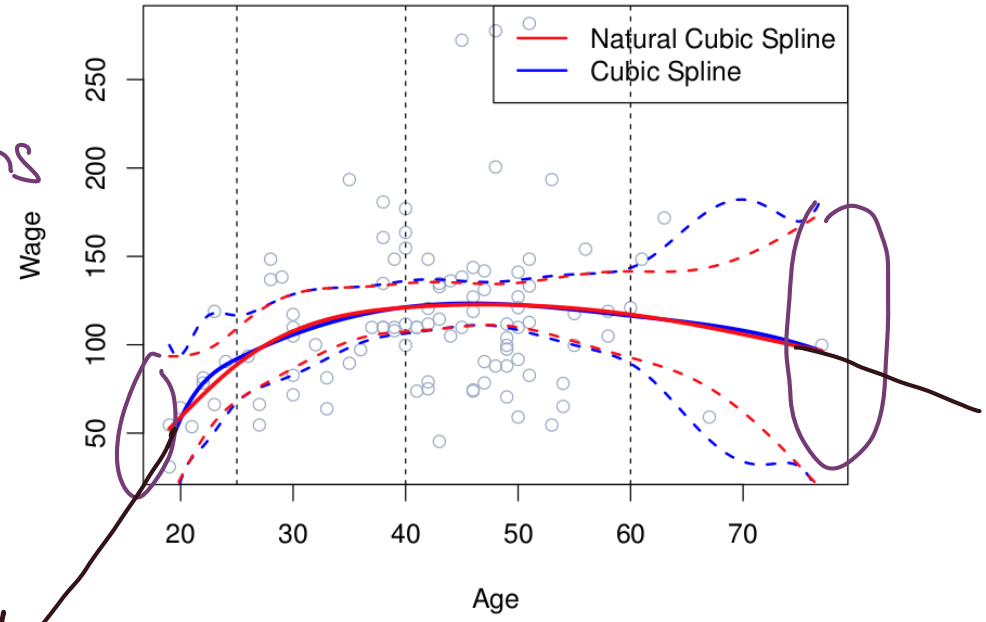
Notes on cubic splines

To human eye, these look like smooth, continuous fcns

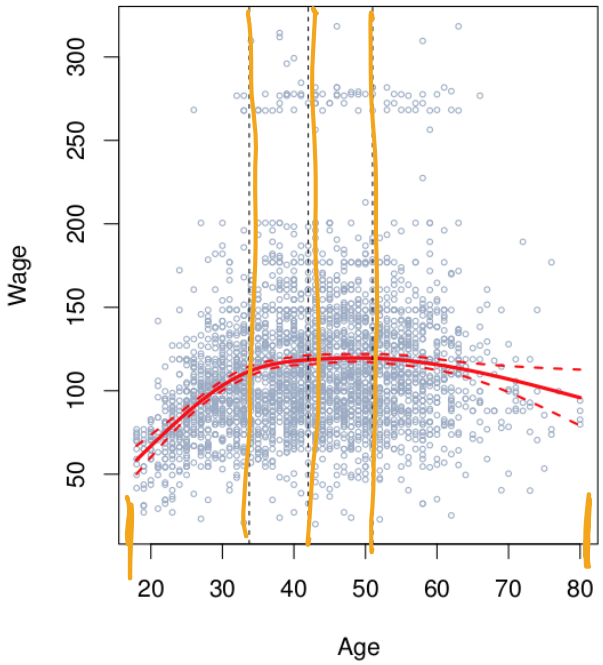
Get wider variance at ends of data set

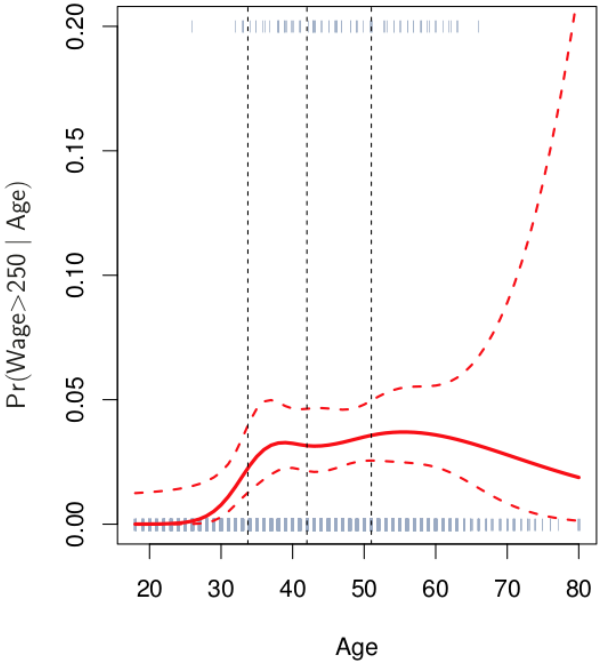
$\boxed{\text{Natural cubic spline}}$

Regression spline plus boundary constraints requiring the fcn is linear



Where to put the knots?





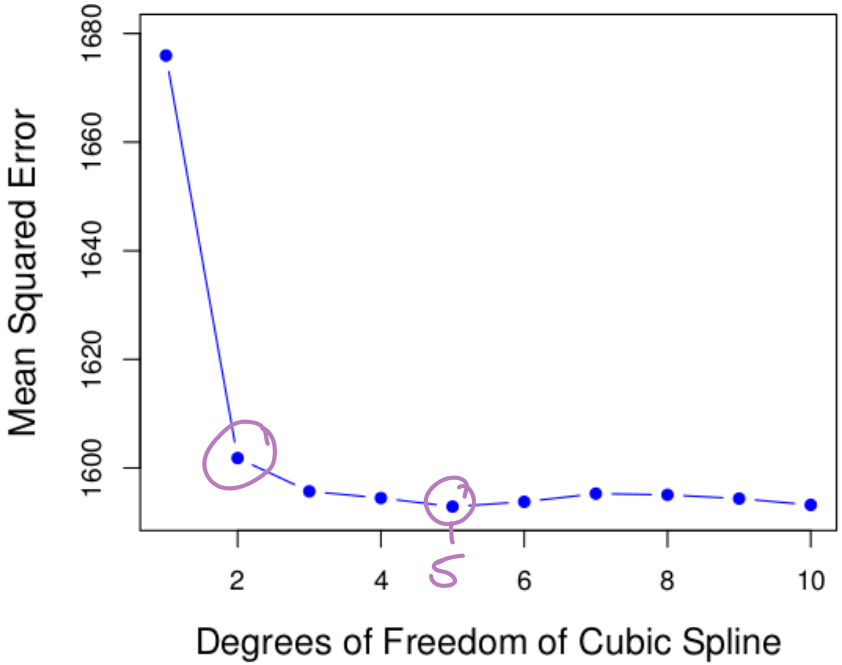
Plan A: Uniform placement (evenly spaced)

Plan B: data driven version

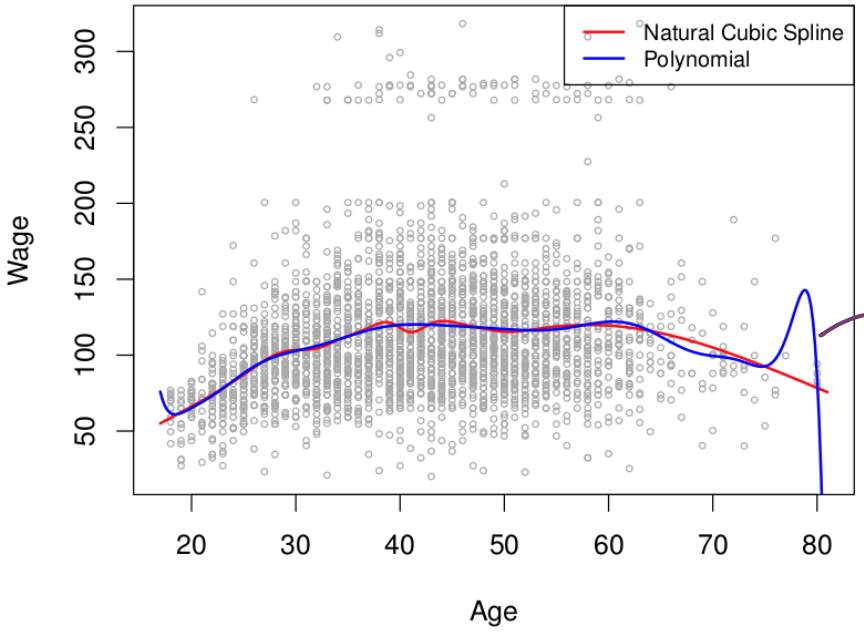
In this ex, put @ the quantiles of data

How many knots to use?

When in doubt, Cross-Validate.



Cubic splines vs Polynomial Regression



polynomial weird boundary behavior

Splines add flexibility with knots rather than poly degree, so get smoother outputs

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